

**Design technology Standard level Paper 2**

Set A

Student Name:

**1 hour 30 minutes**

**Instructions to candidates**

- Write your session number in the boxes above.
- Do not open this examination paper until instructed to do so.
- Answer all questions.
- Answers must be written within the answer boxes provided.
- The maximum mark for the examination paper is [50 marks].

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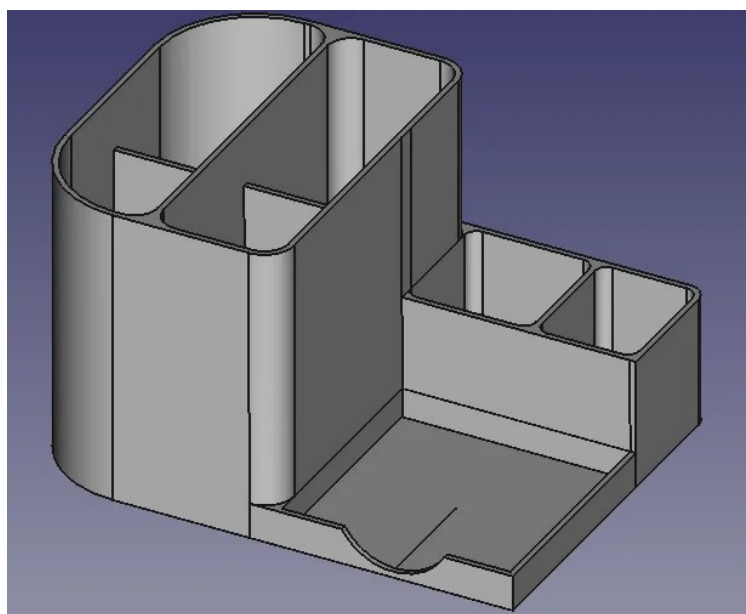
Answer all questions. Answers must be written within the answer boxes provided.

1. A shared grade 12 study room contains four desks used by different students throughout the day. Chargers, cables and power banks are left loose on the desks, and cables are frequently pulled onto the floor.

An IB design technology student designed a desk organiser to hold and route this equipment.

*Table 1: Extract from a student's design project product function specifications*

| Product Function | Essential                                                                                                                                              | Desirable                                                           |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| 1.1              | Free-standing on a shared desk, with a footprint no larger than 200mm x 200mm                                                                          |                                                                     |
| 1.2              | Must hold and route <ul style="list-style-type: none"> <li>• 4 charging cables</li> <li>• 2 power banks</li> <li>• 1 multi-socket extension</li> </ul> |                                                                     |
| 1.3              | Cables must remain accessible without lifting the unit                                                                                                 |                                                                     |
| 1.4              |                                                                                                                                                        | Include a removable tray that one student can carry to another desk |



*Figure 1: A CAD image of a desk organizer. Note that this does not match the described design, Mr. K didn't have time to jump into Fusion to create an accurate image for this question.*

The student's drawings include the following notes:

- "Slot width increased from 6 mm to 11 mm after users could not fit braided USB-C cables"
- "Base weighted after the unit tipped when four cables were pulled at once"
- "Removable tray added after interviews: students move rooms between lessons"
- "Chamfer added at cable exits to reduce wear on cable sheaths"
- "Ventilation slots added below the power bank shelf after the unit reached 41 °C in testing"
- "Footprint reduced from 240 mm to 195 mm square to meet the specification"

*(Question continues on the following page)*

*(Question 1, continued)*

**(a)** State one user-centred research method the student may have used to understand the needs of students using the shared study room, see Table 1 and Figure 1. **[1]**

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**(b)** Describe one improvement to the student's design that resulted from their testing and refinement, see Figure 1. **[2]**

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**(c)** Explain how the essential and desirable criteria in Table 1 guided the development of the student's design, see Table 1 and Figure 1. **[3]**

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**2.** USB (Universal Serial Bus) is a connector and communication standard first published in 1996 by a group of manufacturers who agreed to replace the many incompatible ports then in use on computers.

The original USB-A connector could be inserted only one way round, and its correct orientation was difficult to see. The USB-C connector, published in 2014, is symmetrical, so it can be inserted either way up. It carries power as well as data, and a single USB-C cable can now charge a phone, a laptop and a set of headphones.



*Figure 2: USB-A and USB-C connectors shown to the same scale, with the internal metal contacts visible and the USB-A orientation key indicated*

**(a)** State one physical property that is relevant to the selection of the material used for the metal contacts inside the connector, see Figure 2. **[1]**

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**(b)** State one chemical property that is relevant to the selection of the material used for the metal contacts, see Figure 2. **[1]**

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*(This question continues on the following page)*

*(Question 2 continued)*

**(c)** Outline how the symmetrical USB-C connector improves the product for users with low or no vision, see Figure 2. **[2]**

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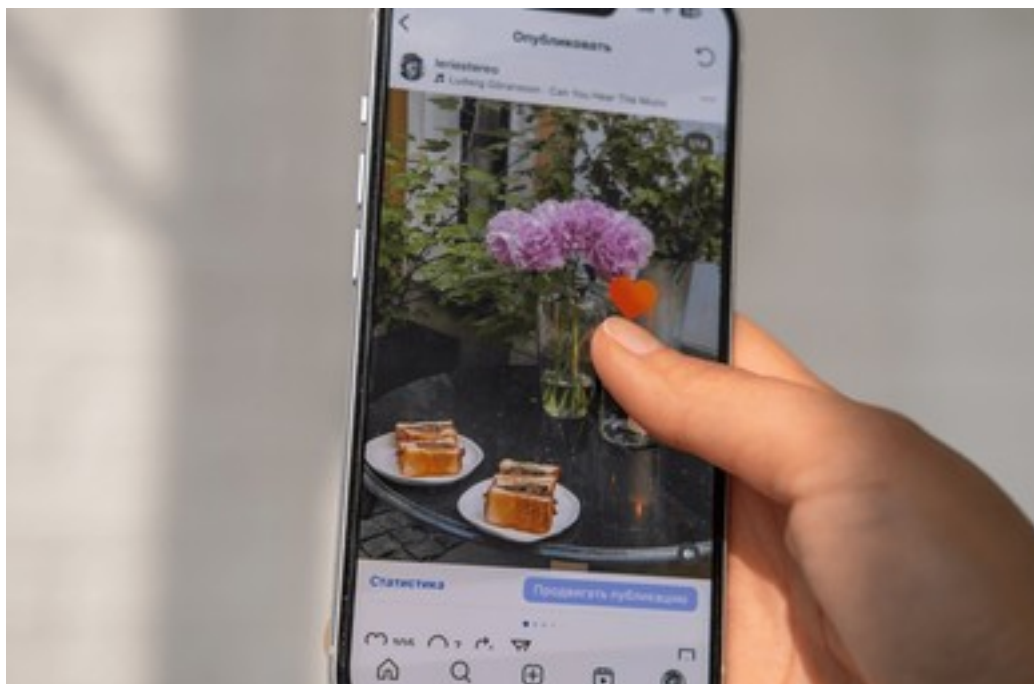
**(d)** Describe how agreement between manufacturers on a single connector standard reduces waste. **[2]**

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**3.** Instagram launched in 2010 as a photo-sharing application for phones. Its home screen presents a single vertical feed, scrolled with one thumb while the phone is held in one hand.

In 2016 the chronological feed was replaced with one ordered by an algorithm. The layout of the feed itself has changed very little since launch.

In 2021, following research into the effects of prolonged use on adolescent wellbeing, the company introduced a "Take a break" reminder that interrupts the feed after a period of continuous scrolling and suggests the user stops for a time.



*Figure 3: The Instagram feed shown in a one-handed grip, with the thumb's arc of movement overlaid.*

**(a)** State the usability objective demonstrated by a returning user finding the feed's layout unchanged after several months away, see Figure 3. **[1]**

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**(b)** Present the reach and work envelope considerations that apply to a user scrolling the feed with one thumb, see Figure 3. **[2]**

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(Question 3 continued)

(c) Evaluate the "Take a break" reminder as a response to the responsibility of the designer, see Figure 3. **[3]**

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4. The Sony TPS-L2, sold from 1979 as the Walkman, was the first widely available personal cassette player. It measured approximately 88 mm × 134 mm × 29 mm, ran on two AA batteries and was carried on a belt clip or in a coat pocket.

It had no recording function and no speaker. Sony's own engineers had argued that a cassette player without a recording function would not sell. It was fitted with two headphone sockets so that two people could listen at once.

More than 200 million cassette Walkmans were sold before the format was displaced.



Figure 4: The Sony TPS-L2 Walkman (belt clip not visible)

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*(Question 4 continued)*

**(a)** State the type of obsolescence that ended the production of cassette Walkmans. **[1]**

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**(b)** Describe why the mechanical properties of the polymer casing were critical to a product designed to be carried on a belt, see Figure 4. **[2]**

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**(c)** Analyse how the dimensions and the belt clip of the TPS-L2 respond to the needs of a user listening while walking, see Figure 4. **[3]**

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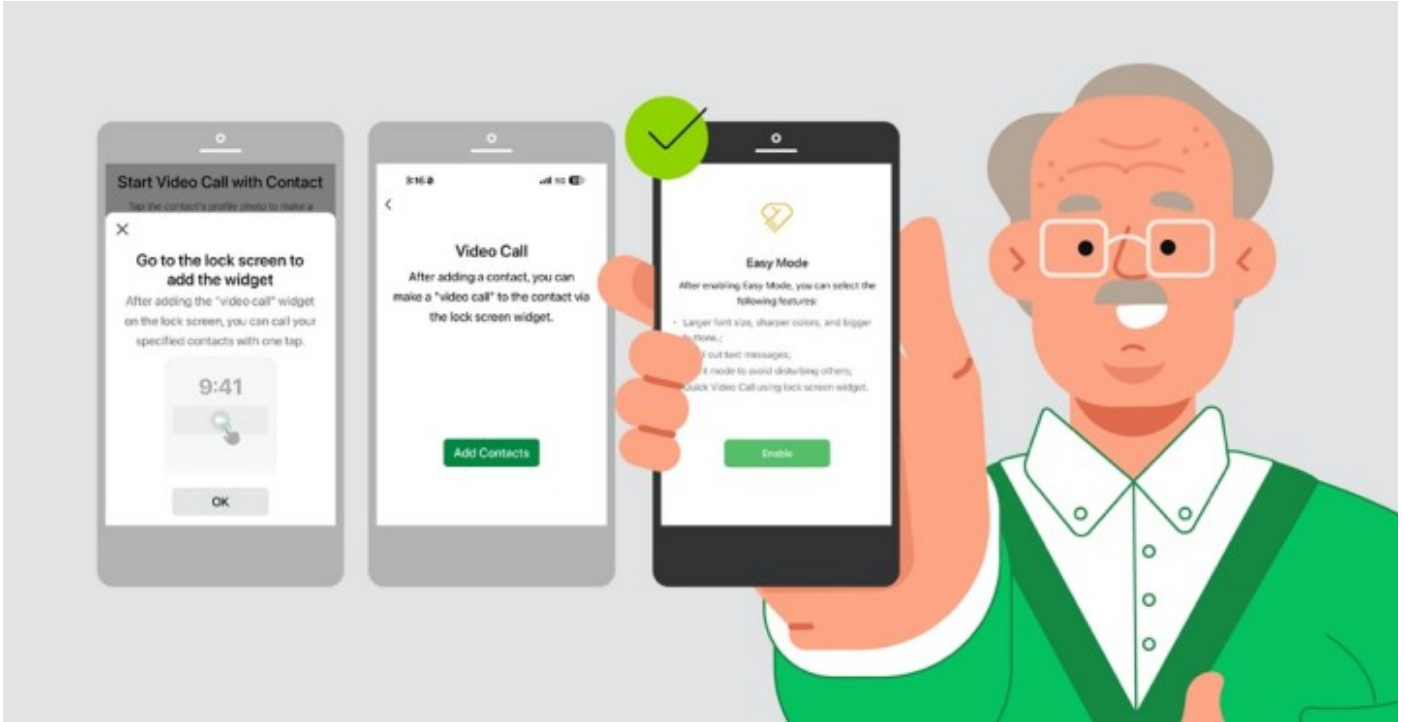
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**5.** WeChat is a messaging application launched in China in 2011. It has grown into a single application used for messaging, payments, ride-hailing, medical appointments, utility bills and government services, with over a billion people using it each month.

Because so many everyday services are now reached only through the application, users who cannot operate it can be shut out of those services altogether.

In 2021 the developer released a "care mode" with larger text, higher contrast and simplified menus, developed after research with users over 60.



*Figure 5: A WeChat graphic showing 'Easy Mode' (Care Mode)*

**(a)** State one user-centred research method that would identify the difficulties experienced by users over 60, see Figure 5. **[1]**

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**(b)** Identify two changes made in care mode that address limiting aspects of user capability, see Figure 5. **[2]**

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(Question 6 continued)

**(c)** Justify why designing WeChat's standard interface for the average user did not meet the needs of the whole user population, see Figure 5. **[3]**

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**6.** A mechanical keyboard uses an individual spring-loaded switch beneath every key, instead of the single rubber membrane sheet used in most office keyboards.

Enthusiast manufacturers sell the keyboard as a set of separable parts: case, plate, printed circuit board, switches and keycaps. Switches sit in sockets and can be pulled out and replaced by hand, without soldering.

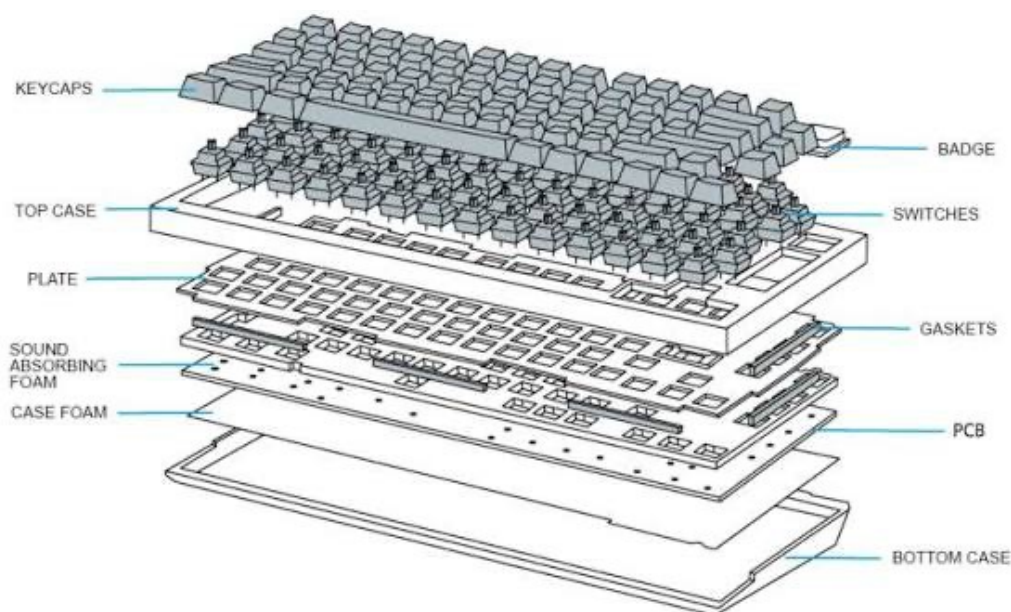


Figure 6: Exploded drawing of a mechanical keyboard, showing (from top) keycaps, switches, plate, printed circuit board and case

**(a)** List two formal drawing techniques, other than the one used in Figure 6, that the manufacturer could use to communicate the assembly of the keyboard. **[2]**

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*(Question 6 continued)*

Keycaps are typically moulded in one of two polymers. ABS is easier to mold into fine shapes, allowing for multi-part keys with elements that allow light to pass through. A downside of ABS is that the material wears down slightly and develops a smooth shine where fingers contact it after a few months of daily use. PBT is harder and more dense, and resists wearing down and developing a shine. It's physical properties make it capable of being printed on with a method that produces high quality, long lasting prints. This printing method is not possible on ABS as the high temperature required causes ABS to warp. PBT is more difficult to mold, requiring more precise temperature control to avoid shrinking or warping. As a result PBT keycaps are harder to produce to a consistent size than ABS. PBT is more durable over time, and is often perceived as a premium material.



*Figure 7: Two keycaps of the same design after two years of daily use, one moulded in ABS and one in PBT.*

**(b)** Outline why a manufacturer would select PBT rather than ABS for keycaps, see Figure 7. [2]

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Typists working at a keyboard for several hours a day report fatigue in the fingers, wrists and forearms. The force needed to actuate a switch varies between products from about 35 g to about 80 g. Some keyboards are split into two halves that can be placed apart on the desk, and tented, so that the inner edge of each half is raised.

*(This question continues on the following page)*

(Question 6 continued)



Figure 8: A conventional keyboard in use (left) and a split, tented mechanical keyboard in use (right)

(c) Describe how the split and tented layout in Figure 8 reduces fatigue for a user typing for several hours. [2]

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(Question 6 continued)

An office keyboard whose rubber membrane has worn out is normally discarded whole. On a mechanical keyboard the switches, keycaps and cable can each be replaced individually, and the case commonly outlasts several sets of components.

**(d)** Explain how the modular construction of a mechanical keyboard supports a circular economy approach, see Figure 6. **[4]**

[illegible]



7. World of Warcraft is an online game first released in 2004 and still running. Players see the game world through an interface of panels showing health, abilities, maps, chat and inventory.

The 2004 interface was fixed. A player could not move, resize or hide any panel. Players responded by writing their own add-on software to rearrange it, and by 2010 most experienced players were running several add-ons in order to play.

In 2022 the developer released an editing mode that allows every panel to be moved, resized and hidden without add-ons.



Figure 9: The 2004 default interface, with ability bar at the bottom and only the player's unit frame.



Figure 10: The 2022 default interface with editing mode active, showing panels which can be repositioned.

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(Question 7 continued)

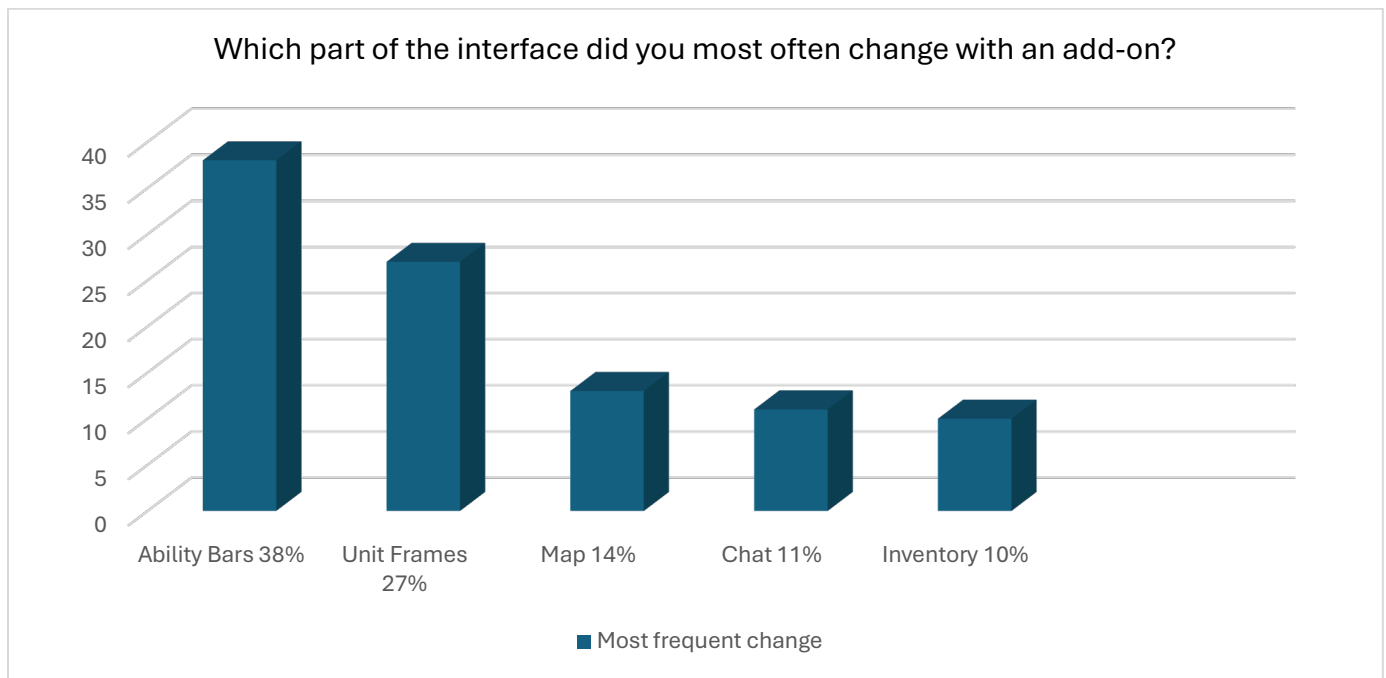


Figure 11: Extract from a player survey with 3412 responses

**(a)** Identify two stakeholders whose feedback would inform a redesign of the interface, see Figure 9 and Figure 10. **[2]**

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**(b)** Outline what the survey data in Figure 11 tells the designers about where the 2004 interface was failing its users. **[2]**

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**(c)** Describe how the 2022 editing mode addresses one of the five usability objectives, see Figure 10. **[2]**

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(Question 7 continued)

**(d)** Evaluate the differences between the 2004 interface and the 2022 interface, see Figures 9–11. **[4]**

[illegible]



# Markscheme

Design technology Standard level Paper 2 Set A

## Marking annotations

The following annotations are available when marking responses.

| Annotation | Meaning                                                                | Annotation | Meaning                      |
|------------|------------------------------------------------------------------------|------------|------------------------------|
| ✓          | Correct point (awards 1 mark, up to the maximum for the question part) | NAQ        | Not answered question        |
| X          | Incorrect point                                                        | NBOD       | No benefit of the doubt      |
| 0          | Incorrect response, awards zero                                        | OC         | Off course                   |
| A+         | Analysis                                                               | REP        | Repetition                   |
| B+         | Balanced argument                                                      | SEEN       | Point noted, no credit given |
| BOD        | Benefit of the doubt                                                   | TV         | Too vague                    |
| D          | Description                                                            | WTTE       | Words to that effect         |
| EVAL       | Evaluation                                                             | ^          | Caret, omission              |
| GEXP       | Good explanation                                                       |            |                              |

Make sure you have looked at all pages. Put the SEEN annotation on any blank page to indicate that you have seen it.

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| Question | Answers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Notes                                                                                                                                                                                                                                                          | Total |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1.       | <p>a</p> <ul style="list-style-type: none"> <li>• Interviews ✓</li> <li>• User observation ✓</li> <li>• Surveys / questionnaires ✓</li> <li>• Focus groups ✓</li> <li>• Field research ✓</li> <li>• Task analysis ✓</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Award [1] for one relevant user-centred research method that may have been used to understand the needs of students using the shared study room up to [1 max].</p>                                                                                          | 1     |
| 1.       | <p>b</p> <ul style="list-style-type: none"> <li>• Slot width increased from 6 mm to 11 mm ✓ because users could not fit braided USB-C cables into the original slot ✓</li> <li>• Base weighted ✓ because the unit tipped when four cables were pulled at once ✓</li> <li>• Removable tray added ✓ in response to interview data showing students move between rooms ✓</li> <li>• Chamfer added at the cable exits ✓ to reduce wear on the cable sheaths ✓</li> <li>• Ventilation slots added below the power bank shelf ✓ after the unit reached 41 °C in testing ✓</li> <li>• Footprint reduced from 240 mm to 195 mm square ✓ to meet the essential specification of 200 mm × 200 mm ✓</li> </ul>                                                                                                                                                                                                                                                                                                                                                                            | <p>Award [1] for each detail, leading to an account of one improvement resulting from testing and refinement up to [2 max].</p> <p>Credit the change and the reason for it. Do not award [2] for naming two changes with no reason.</p>                        | 2     |
| 1.       | <p>c</p> <p>Design specifications communicate the essential and desirable success criteria of the redesigned product [definition].</p> <ul style="list-style-type: none"> <li>• Essential criteria set the limits the design must meet, so the footprint was reduced from 240 mm to 195 mm square to satisfy criterion 1.1 ✓</li> <li>• Criterion 1.2 fixed the quantity and type of items to be held, which determined the number and size of the slots and shelves ✓</li> <li>• Criterion 1.3 required cables to stay accessible without lifting the unit, which drove the open-slot arrangement rather than an enclosed box ✓</li> <li>• Desirable criteria were addressed only once the essential ones were met, so the removable tray in 1.4 was added later in development ✓</li> <li>• Separating essential from desirable allowed the student to prioritize when the two conflicted, for example between a small footprint and a removable tray ✓</li> <li>• Criteria gave the student a measurable standard against which each iteration could be judged ✓</li> </ul> | <p>Award [1] for each relevant reason / cause / piece of evidence showing how the essential and desirable criteria guided the development of the design up to [3 max].</p> <p>Award a maximum of [2] where the response refers only to essential criteria.</p> | 3     |
| 2.       | <p>a</p> <ul style="list-style-type: none"> <li>• Electrical conductivity ✓</li> <li>• Thermal conductivity ✓</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>Award [1] for one relevant physical property up to [1 max].</p>                                                                                                                                                                                             | 1     |

| Question | Answers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Notes                                                                                                                                                                                                     | Total |
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|          | <ul style="list-style-type: none"> <li>• Melting point ✓</li> <li>• Density ✓</li> <li>• Electrical resistivity ✓</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Do not credit mechanical or chemical properties.                                                                                                                                                          |       |
| 2.       | <p><b>b</b></p> <ul style="list-style-type: none"> <li>• Corrosion resistance ✓</li> <li>• Reactivity ✓</li> <li>• Oxidation resistance ✓</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>Award [1] for one relevant chemical property up to [1 max].</p> <p>Corrosion resistance is the expected answer: contacts that oxidize lose conductivity.</p>                                           | 1     |
| 2.       | <p><b>c</b></p> <p>Inclusive design ensures products address the needs of the widest possible audience, regardless of age or ability. This may be achieved because:</p> <ul style="list-style-type: none"> <li>• The connector cannot be inserted the wrong way up, so the user does not need to see the orientation ✓</li> <li>• The user does not need to look at the port before connecting ✓</li> <li>• Errors and repeated attempts are eliminated ✓</li> <li>• The same cable end works in either orientation, reducing the memory burden ✓</li> <li>• The design removes a task that previously depended on visual accuracy ✓</li> <li>• It benefits all users in poor light, not only users with low vision ✓</li> </ul> | <p>Award [1] for each relevant detail identified relating to how the symmetrical USB-C connector supports users with low or no vision up to [2 max].</p>                                                  | 2     |
| 2.       | <p><b>d</b></p> <ul style="list-style-type: none"> <li>• One standard means one cable type serves many products, so users need fewer cables ✓</li> <li>• Cables and chargers are not discarded when a device is replaced ✓</li> <li>• Existing chargers can be reused with new products, extending their useful life ✓</li> <li>• Manufacturers do not need to supply a charger with every product ✓</li> <li>• Fewer proprietary connectors reach the waste stream ✓</li> <li>• Reduced material and energy consumption across the whole product category ✓</li> </ul>                                                                                                                                                          | <p>Award [1] for each detail considered, leading to a balanced account of how a single connector standard reduces waste up to [2 max].</p>                                                                | 2     |
| 3.       | <p><b>a</b></p> <ul style="list-style-type: none"> <li>• Memorability ✓</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Award [1] for the correct usability objective up to [1 max].</p> <p>The five usability objectives are learnability, efficiency, memorability, errors and satisfaction. Do not credit learnability.</p> | 1     |

| Question | Answers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Notes                                                                                                                                                                                                                                                                 | Total |
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| 3.       | <p>Consideration must be given to work envelopes, reach, clearance, adjustability and range of sizes.</p> <ul style="list-style-type: none"> <li>• The thumb sweeps an arc, not a straight line, so the reachable area is a curved band rather than the whole screen ✓</li> <li>• Controls placed at the top of the screen fall outside the thumb's work envelope in a one-handed grip ✓</li> <li>• The feed is scrolled with a vertical swipe, which sits within the thumb's natural arc ✓</li> <li>• Reach varies with hand size, so a control comfortable for a 95th percentile hand may be unreachable for a 5th percentile hand ✓</li> <li>• A single vertical feed removes the need to reach across the screen at all ✓</li> </ul>                                                                                                                                                                                                                                                    | <p>Award [1] for each relevant reach or work envelope consideration presented for a one-handed grip up to [2 max].</p>                                                                                                                                                | 2     |
| 3.       | <p>A designer has a responsibility to the needs of clients, their community and the environment. It is a designer's responsibility to ensure their products are safe to use.</p> <p>Strengths:</p> <ul style="list-style-type: none"> <li>• Responds directly to evidence of harm identified by the company's own research ✓</li> <li>• Places a limit inside the product rather than relying on the user to self-regulate ✓</li> <li>• Addresses wellbeing, which extends the designer's duty of safety beyond physical harm ✓</li> </ul> <p>Limitations:</p> <ul style="list-style-type: none"> <li>• The reminder can be dismissed, so it does not change the underlying design ✓</li> <li>• The feed remains algorithmically ordered to maximize time spent, which conflicts with the reminder's purpose ✓</li> <li>• It places the responsibility back on the user rather than removing the cause ✓</li> <li>• It was introduced years after the behaviour was identified ✓</li> </ul> | <p>Award [1] for each distinct strength / limitation, leading to an appraisal of the "Take a break" reminder as a response to the responsibility of the designer up to [3 max].</p> <p>Award a maximum of [2] where only strengths or only limitations are given.</p> | 3     |
| 4.       | <p>a</p> <ul style="list-style-type: none"> <li>• Technological obsolescence ✓</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Award [1] for the correct type of obsolescence up to [1 max].</p> <p>Do not credit style, social, functional or planned obsolescence.</p>                                                                                                                          | 1     |
| 4.       | <p>b</p> <p>Mechanical properties include aspects of a material affected by the application of a force.</p> <ul style="list-style-type: none"> <li>• Toughness, so that the casing absorbs the impact of being dropped without fracturing ✓</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Award [1] for each detail of a relevant mechanical property and its consequence for a belt-carried product, leading to an</p>                                                                                                                                      | 2     |

| Question | Answers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Notes                                                                                                                                                                                                                                                                                                      | Total |
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|          | <ul style="list-style-type: none"> <li>• Hardness, to resist scratching against a belt, coat or bag ✓</li> <li>• Stiffness, so the casing does not flex and disturb the tape transport mechanism ✓</li> <li>• Elasticity, so the casing returns to shape after being knocked ✓</li> <li>• A brittle casing would crack at the belt clip, which carries the whole weight of the product ✓</li> <li>• Low density keeps the product light enough to be worn ✓</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>account up to [2 max].</p> <p>Density is a physical property; credit it only where the response also names a mechanical property.</p>                                                                                                                                                                   |       |
| 4.       | <p>c</p> <p>Anthropometric data is used to design for different people; consideration must be given to reach, clearance and range of sizes.</p> <ul style="list-style-type: none"> <li>• The 88 mm × 134 mm footprint is close to the dimensions of an adult hand, so the player can be held and operated one-handed ✓</li> <li>• The 29 mm depth keeps the product flat enough to sit against the body on a belt without obstructing the hip ✓</li> <li>• Belt mounting moves the product's mass to the waist rather than the hand, so the user's hands remain free while walking ✓</li> <li>• Clearance between the belt clip and the casing must suit the range of belt thicknesses in use ✓</li> <li>• Controls must fall within reach of the fingers of the hand holding the unit, or be operable without looking when the unit is worn ✓</li> <li>• A dynamic anthropometric consideration: the product is operated while the body is in motion, so controls must tolerate imprecise finger placement ✓</li> </ul> | <p>Award [1] for each guiding element / structure identified relating to how the dimensions and belt clip respond to the needs of a user listening while walking up to [3 max].</p> <p>Award a maximum of [2] where the response addresses only static dimensions and does not consider use in motion.</p> | 3     |
| 5.       | <p>a</p> <ul style="list-style-type: none"> <li>• Interviews ✓</li> <li>• User observation ✓</li> <li>• Focus groups ✓</li> <li>• Surveys / questionnaires ✓</li> <li>• Field research ✓</li> <li>• Task analysis ✓</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>Award [1] for one relevant user-centred research method that would identify the difficulties experienced by users over 60 up to [1 max].</p>                                                                                                                                                            | 1     |
| 5.       | <p>b</p> <p>Students must be able to explain limiting aspects of user capabilities, including visual accuracy, colour perception, strength, fatigue, muscle control and hearing thresholds.</p> <ul style="list-style-type: none"> <li>• Larger text addresses reduced visual accuracy ✓</li> <li>• Higher contrast addresses reduced colour perception and contrast sensitivity ✓</li> <li>• Simplified menus reduce the memory burden</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>Award [1] for each relevant change identified and the limiting capability it addresses, up to [2 max].</p> <p>Do not award a mark for naming a change with no reference to a user capability.</p>                                                                                                       | 2     |

| Question | Answers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Notes                                                                                                                                                                                                                                                                                       | Total |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|          | <p>and the number of decisions ✓</p> <ul style="list-style-type: none"> <li>• Larger touch targets accommodate reduced fine muscle control ✓</li> <li>• Fewer items per screen reduces visual search time and fatigue ✓</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                             |       |
| 5.       | <p>Inclusive design is not always possible. The average person correlates to the 50th percentile adult, and it is not always appropriate to design for the average person. Valid reasons and evidence:</p> <ul style="list-style-type: none"> <li>• Designing for the 50th percentile excludes users at both extremes of a capability range ✓</li> <li>• Visual accuracy and contrast sensitivity decline with age, so an interface set for an average adult is unusable for a substantial older population ✓</li> <li>• WeChat's user population is over a billion people and is not homogeneous, so no single average describes it ✓</li> <li>• The consequence of exclusion is unusually severe here, because payments, medical appointments and government services are reachable only through the application ✓</li> <li>• Excluded users are shut out of essential services, not merely inconvenienced ✓</li> <li>• A design-for-extremes strategy, adopted in care mode, produces a solution that also benefits average users ✓</li> <li>• The company's own research with users over 60 provided the evidence that the standard interface was failing ✓</li> </ul> | <p>Award [1] for each valid reason / piece of evidence supporting why designing for the average user did not meet the needs of the whole user population up to [3 max].</p> <p>Credit responses that reason from the 50th percentile, from the consequences of exclusion, or from both.</p> | 3     |
| 6.       | <p>Students must be able to construct and interpret 2D drawings and 3D models. Figure 6 is an exploded drawing, so do not credit it.</p> <ul style="list-style-type: none"> <li>• Isometric / assembled drawing ✓</li> <li>• Orthographic projection ✓</li> <li>• Sectioned drawing ✓</li> <li>• Perspective drawing ✓</li> <li>• Detail or part drawing with dimensions ✓</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Award [1] for each relevant formal drawing technique up to [2 max].</p> <p>Do not credit exploded drawing. Do not credit informal techniques such as freehand sketching.</p>                                                                                                             | 2     |
| 6.       | <p>Materials are selected based on their properties and on aesthetic characteristics including texture, form and colour.</p> <ul style="list-style-type: none"> <li>• PBT resists the shine that develops on ABS, so the keycap keeps its appearance over years of use ✓</li> <li>• Greater surface hardness gives better wear resistance at the finger contact point ✓</li> <li>• The slightly textured surface improves grip</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>Award [1] for each relevant brief point relating to why PBT is selected in preference to ABS up to [2 max].</p> <p>Credit a response that identifies the trade-off as one of its points.</p>                                                                                             | 2     |

| Question | Answers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Notes                                                                                                                                                                                                                                                                                                                       | Total |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|          | <p>and finger location ✓</p> <ul style="list-style-type: none"> <li>• Longer useful life reduces the need to replace keycaps ✓</li> <li>• The trade-off is greater shrinkage during moulding, so tolerances are harder to hold and unit cost is higher ✓</li> <li>• PBT may be printed with a method that is not possible with ABS ✓</li> <li>• A company may choose to use PBT due to market demand as a result of perceived superiority ✓</li> </ul> <p>Physiology is the study of systems and biomechanics within the human body, their responses, limitations and capabilities.</p> <ul style="list-style-type: none"> <li>• Splitting the halves lets each hand sit in line with the shoulder, removing the sideways bend of the wrist ✓</li> <li>• Tenting raises the inner edges so the forearms are not twisted flat against the desk ✓</li> </ul> |                                                                                                                                                                                                                                                                                                                             |       |
| 6.       | <p>c</p> <ul style="list-style-type: none"> <li>• A more neutral wrist and forearm position reduces sustained static muscle load ✓</li> <li>• Reduced static loading over several hours delays the onset of fatigue ✓</li> <li>• The user can set the separation to suit their own shoulder width rather than a fixed average ✓</li> <li>• Lower actuation force, from about 35 g, further reduces the work done by each finger over a working day ✓</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                            | <p>Award [1] for each detail of how the split and tented layout reduces fatigue, leading to an account up to [2 max].</p>                                                                                                                                                                                                   | 2     |
| 6.       | <p>d</p> <p>A circular economy is a closed-loop system where resources are continuously repurposed. Designers can eliminate waste through longevity, upgradability, disassembly and dematerialization, and can recover value through reuse, repair, recondition and recycling.</p> <p>Longevity and disassembly:</p> <ul style="list-style-type: none"> <li>• The keyboard separates into case, plate, printed circuit board, switches and keycaps, so it is designed for disassembly ✓</li> <li>• No soldering is needed to remove a switch, so disassembly requires no specialist skill or equipment ✓</li> <li>• The case commonly outlasts several sets of components, extending the life of the largest single part ✓</li> </ul> <p>Repair:</p> <ul style="list-style-type: none"> <li>• A single failed switch is replaced rather than</li> </ul>    | <p>Award [1] for each relevant detail / reason / cause relating to how the modular construction of the keyboard supports a circular economy approach up to [4 max].</p> <p>A response confined to repair alone should be awarded a maximum of [3]. Credit responses that reach the comparison with a membrane keyboard.</p> | 4     |



| Question | Answers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Notes                                                                                                                                                                                                                                                                     | Total |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|          | <p>the whole product ✓</p> <ul style="list-style-type: none"> <li>• Worn keycaps are replaced without discarding the keyboard ✓</li> <li>• A damaged cable is replaced independently ✓</li> </ul> <p>Comparison with the membrane keyboard:</p> <ul style="list-style-type: none"> <li>• A worn membrane sheet cannot be replaced, so the whole product is discarded ✓</li> <li>• The mechanical keyboard therefore keeps materials in use for far longer ✓</li> </ul> <p>Upgradability and reuse:</p> <ul style="list-style-type: none"> <li>• Switches can be changed for a different actuation force without replacing the keyboard, which is an upgrade rather than a replacement ✓</li> <li>• Removed components remain usable and can be reused in another build ✓</li> </ul> <p>Limits:</p> <ul style="list-style-type: none"> <li>• Sold as separate parts, the product uses more packaging and more transport ✓</li> <li>• Replaceability only reduces waste if replacement parts remain available ✓</li> </ul> |                                                                                                                                                                                                                                                                           |       |
| 7.       | <p><b>a</b></p> <p>Students must be able to identify the various stakeholders, such as users, manufacturers and engineers, that will help in any future development of a product.</p> <ul style="list-style-type: none"> <li>• Players / users ✓</li> <li>• New players, as a distinct user group from experienced players ✓</li> <li>• Add-on developers ✓</li> <li>• The game's own interface designers and engineers ✓</li> <li>• Community managers ✓</li> <li>• Accessibility specialists ✓</li> <li>• The publisher, as the commercial stakeholder ✓</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Award [1] for each relevant stakeholder identified up to [2 max].</p> <p>Do not credit two descriptions of the same stakeholder group.</p>                                                                                                                             | 2     |
| 7.       | <p><b>b</b></p> <ul style="list-style-type: none"> <li>• The most-changed elements were the ability bars at 38 % and the unit frames at 27 % ✓</li> <li>• Together these two account for 65 % of add-on use, so failure was concentrated in a small part of the interface rather than spread across it ✓</li> <li>• The elements changed most often are the ones used most often in play, so the interface failed hardest where it mattered most ✓</li> <li>• Map, chat and inventory each account for less than 15 %, so those panels were broadly acceptable ✓</li> <li>• That players wrote software to fix the</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                            | <p>Award [1] for each relevant brief point summarizing what the survey data indicates about where the 2004 interface failed its users up to [2 max].</p> <p>Award a maximum of [1] where the response only restates figures from the chart without interpreting them.</p> | 2     |

| Question | Answers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Total             |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
|          | <p>interface themselves is evidence of unmet need ✓</p> <ul style="list-style-type: none"> <li>• The sample of 3412 responses is large enough to indicate a population-level problem rather than individual preference ✓</li> </ul> <p>The five usability objectives are learnability, efficiency, memorability, errors and satisfaction.</p> <p>Efficiency:</p> <ul style="list-style-type: none"> <li>• Frequently used panels can be moved within immediate view, reducing the time spent searching ✓</li> </ul> <p>Errors:</p> <ul style="list-style-type: none"> <li>• Panels can be repositioned away from each other so the wrong control is less likely to be selected ✓</li> </ul> <p>Satisfaction:</p> <ul style="list-style-type: none"> <li>• The interface can be set to individual preference, so users are more satisfied with it ✓</li> </ul> <p>Learnability:</p> <ul style="list-style-type: none"> <li>• A new player no longer has to find, install and configure add-ons before the interface becomes usable ✓</li> </ul> <p>Memorability:</p> <ul style="list-style-type: none"> <li>• A layout the user arranged themselves is easier to recall after time away ✓</li> </ul> <p>Strengths of the 2022 interface:</p> <ul style="list-style-type: none"> <li>• Every panel can be moved, resized and hidden without third-party software ✓</li> <li>• The functionality players had built for themselves is now supported and maintained by the developer ✓</li> <li>• New players get a usable interface without installing anything ✓</li> <li>• Removes the risk of add-ons breaking when the game updates ✓</li> </ul> <p>Limitations of the 2022 interface:</p> <ul style="list-style-type: none"> <li>• A player must still configure it themselves, so the default layout is still not right for everyone ✓</li> <li>• Editing mode adds a layer of choice that can overwhelm a new player ✓</li> <li>• Some add-on functions remain outside what the built-in mode can do ✓</li> </ul> <p>Weaknesses of the 2004 interface:</p> | <p>Award [1] for each detail describing how the editing mode addresses one named usability objective, leading to an account up to [2 max].</p> <p>The response must name the objective. Award a maximum of [1] where the objective is described but not named.</p> <p>Award [1] for each distinct strength / limitation, leading to an appraisal of the differences between the 2004 and the 2022 interface up to [4 max].</p> <p>Award a maximum of [3] where the response does not weigh the two against each other. Credit responses that use Figure 11 as evidence.</p> | <p>2</p> <p>4</p> |
| 7.       | c                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                   |
| 7.       | d                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                   |

| Question | Answers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Notes | Total |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|
|          | <ul style="list-style-type: none"> <li>• Fixed panels could not be adapted to any individual, screen size or play style ✓</li> <li>• Players were required to solve the developer's design problem themselves ✓</li> <li>• Dependence on add-ons excluded players unable or unwilling to install them ✓</li> </ul> <p>Judgment:</p> <ul style="list-style-type: none"> <li>• Eighteen years elapsed between the problem being evident and the developer addressing it ✓</li> <li>• The redesign is a response to demonstrated user behaviour rather than to stated preference, which is stronger evidence ✓</li> </ul> |       |       |